

V7 Method Inventory

This is the planned method inventory for your project after V7 refactoring.

Main Class

Program Startup / Control

main(String[] args)

Purpose:

Start the program.

Responsible for:

Calling runProgram()

Not responsible for:

Menus

Input validation

Task logic

Saving/loading details

runProgram()

Purpose:

Control the main program loop.

Responsible for:

Loading tasks

Showing the main menu

Getting menu choice

Calling handleChoice()

Saving tasks before exit

Not responsible for:

Adding tasks directly
Deleting tasks directly
Searching/filtering/sorting directly

Main Menu

`showMainMenu()`

Purpose:

Display the main menu.

`getMainMenuChoice()`

Purpose:

Get a valid main menu choice.

Uses:

`getValidatedInt()`

`handleChoice(int choice)`

Purpose:

Route the user to the correct handler.

Should use:

`switch(choice)`

Not responsible for:

Doing task actions directly
Long workflows
Validation logic

Handler Methods

handleAddTask()

Purpose:

Coordinate adding a task.

Flow:

getTaskName()
manager.addTask()
displaySuccess()

handleViewTasks()

Purpose:

Display all tasks.

Flow:

manager.getTasks()
displayTasks()

handleSearchTasks()

Purpose:

Coordinate searching tasks.

Flow:

getSearchTerm()
manager.searchTasks()
displayTasks()

handleFilterTasks()

Purpose:

Coordinate filtering tasks.

Flow:

showFilterMenu()
getFilterChoice()
call correct filter helper

displayTasks()

handleSortTasks()

Purpose:

Coordinate sorting tasks.

Flow:

showSortMenu()

getSortChoice()

manager.sortByPriority() OR manager.sortByCompletionStatus()

displayTasks()

handleCompleteTask()

Purpose:

Coordinate marking a task complete.

Flow:

getTaskNumber()

manager.completeTask()

displaySuccess()

handleDeleteTask()

Purpose:

Coordinate deleting a task.

Flow:

getTaskNumber()

manager.deleteTask()

displaySuccess()

handleRenameTask()

Purpose:

Coordinate renaming a task.

Flow:

```
getTaskNumber()  
getTaskName()  
manager.renameTask()  
displaySuccess()
```

handleMarkIncomplete()

Purpose:

Coordinate marking task incomplete.

Flow:

```
getTaskNumber()  
manager.markTaskIncomplete()  
displaySuccess()
```

handleSetPriority()

Purpose:

Coordinate setting priority.

Flow:

```
getTaskNumber()  
getPriorityChoice()  
getPriorityFromChoice()  
manager.setTaskPriority()  
displaySuccess()
```

handleSetCategory()

Purpose:

Coordinate setting category.

Flow:

```
getTaskNumber()  
getCategoryChoice()  
getCategoryFromChoice()  
manager.setTaskCategory()
```

displaySuccess()

handleExit()

Purpose:

Exit the program safely.

Flow:

saveTasks()

display exit message

programRunning = false

Input Methods

getTaskName()

Purpose:

Get a valid task name.

Validation:

Cannot be empty

Cannot contain comma

0 can return to menu if used during rename/add

getTaskNumber()

Purpose:

Get a valid task number.

Validation:

Must be between 1 and manager.sizeOfList()

0 = go back

Letters rejected

Negative numbers rejected

getSearchTerm()

Purpose:

Get search text.

Validation:

Cannot be blank

0 = go back

getPriorityChoice()

Purpose:

Get priority menu choice.

Valid choices:

0 = go back

1 = High

2 = Medium

3 = Low

getCategoryChoice()

Purpose:

Get category menu choice.

Valid choices:

0 = go back

1 = School

2 = Work

3 = Personal

4 = Fitness

5 = Church

6 = Other

getFilterChoice()

Purpose:

Get filter menu choice.

Valid choices:

0 = go back
1 = Priority
2 = Category
3 = Completion Status

getSortChoice()

Purpose:

Get sort menu choice.

Valid choices:

0 = go back
1 = Priority
2 = Completion Status

Menu Display Methods

showPriorityMenu()

Displays:

1. High
 2. Medium
 3. Low
 0. Return
-

showCategoryMenu()

Displays:

1. School
 2. Work
 3. Personal
 4. Fitness
 5. Church
 6. Other
 0. Return
-

`showFilterMenu()`

Displays:

1. Filter by Priority
 2. Filter by Category
 3. Filter by Completion Status
 0. Return
-

`showSortMenu()`

Displays:

1. Sort by Priority
 2. Sort by Completion Status
 0. Return
-

Conversion Methods

`getPriorityFromChoice(int choice)`

Purpose:

Convert priority number to String.

- 1 → High
 - 2 → Medium
 - 3 → Low
-

`getCategoryFromChoice(int choice)`

Purpose:

Convert category number to String.

- 1 → School
 - 2 → Work
 - 3 → Personal
 - 4 → Fitness
 - 5 → Church
 - 6 → Other
-

Validation Helpers

getValidatedInt(String prompt, String errorMessage, int min, int max, boolean allowZero)

Purpose:

Reusable integer validation.

Responsible for:

Displaying prompt

Rejecting letters

Rejecting numbers outside range

Allowing 0 if allowZero is true

Returning valid number

isValidTaskNumber(int taskNumber)

Purpose:

Check whether task number exists.

taskNumber >= 1 && taskNumber <= manager.sizeOfList()

Display Helpers

displayTasks(ArrayList<Task> tasks)

Purpose:

Display numbered task lists.

Used for:

View

Search results

Filter results

Sort results

displayHeader(String title)

Purpose:

Display section title.

displaySuccess(String message)**Purpose:**

Display success message.

displayError(String message)**Purpose:**

Display error message.

Save / Load Helpers

loadTasks()**Purpose:**

Ask FileManager to load saved tasks.

saveTasks()**Purpose:**

Ask FileManager to save current tasks.

Important:

Main decides WHEN to save.

FileManager knows HOW to save.

TaskManager Class

Core Task Methods

`addTask(String name)`

Adds a new task.

`completeTask(int taskNumber)`

Marks task complete.

`markTaskIncomplete(int taskNumber)`

Marks task incomplete.

`deleteTask(int taskNumber)`

Deletes task.

`renameTask(int taskNumber, String newName)`

Renames task.

`setTaskPriority(int taskNumber, String priority)`

Updates task priority.

`setTaskCategory(int taskNumber, String category)`

Updates task category.

TaskManager Query Methods

`hasTask()`

Returns whether task list is not empty.

`sizeOfList()`

Returns number of tasks.

`getTaskByNumber(int taskNumber)`

Returns task by user-facing task number.

`getTasks()`

Returns task list.

TaskManager Search / Filter / Sort

`searchTasks(String searchTerm)`

Returns matching tasks by name.

`filterByPriority(String priority)`

Returns tasks with matching priority.

`filterByCategory(String category)`

Returns tasks with matching category.

filterByCompletionStatus(boolean completed)

Returns completed or incomplete tasks.

sortByPriority()

Returns copied list sorted:

High
Medium
Low

sortByCompletionStatus()

Returns copied list sorted:

Incomplete
Completed

TaskManager Private Helpers

getPriorityValue(String priority)

Converts priority to sorting value.

High → 1
Medium → 2
Low → 3
Unknown → 4

Task Class

Constructors

Task(String name)

Used for new tasks.

Defaults:

completed = false

priority = Medium

category = Other

Task(String name, boolean completed, String priority, String category)

Used when loading from file.

Task Getters

getName()

isCompleted()

getPriority()

getCategory()

Task Setters / Mutators

changeName(String newName)

markCompleted()

markIncomplete()

setPriority(String priority)

setCategory(String category)

Task Display

toString()

Returns:

Task Name [] (Priority) [Category]

or:

Task Name [X] (Priority) [Category]

FileManager Class

FileManager(String fileName)

Stores file name.

saveTasks(ArrayList<Task> tasks)

Saves tasks to file.

loadTasks()

Loads tasks from file.